Dhana Karthikeya Ventrapragada

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Education

University of California, Berkeley **Expected Dec 2027**

Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00 *Berkeley, CA*

- **Relevant Coursework:** Foundations of Data Science, Principles & Techniques of Data Science, Probability for Data Science; *In progress:* Introduction to Machine Learning, Data Engineering

Technical Skills

Languages: Python, SQL, Java, TypeScript

Statistics & ML: A/B Testing, Hypothesis Testing, Bootstrapping, scikit-learn, XGBoost, PyTorch, Cross-Validation

Data: pandas, NumPy, Matplotlib, seaborn, PostgreSQL (query plans, indexing), Plotly, Feature Engineering, Jupyter

Engineering: FastAPI, React, Supabase, Claude API, REST APIs, AWS S3, pytest, Git

Experience

IDX Exchange | Data Science Intern | Remote **Jun 2026 - Aug 2026**

- Developed a machine learning model that predicts California home sale prices, typically within **8.6%** of the actual price, using **71K** single-family sales cleaned from **143K** raw records.
- Benchmarked four model types and selected a tuned XGBoost (R^2 of **0.88**), finding that deeper trees improved accuracy more than a lower learning rate.
- Resolved a data leakage bug that had invalidated early results, and nearly halved a simpler model's error (from **29%** to **15%**) by engineering a school district feature.
- Analyzed errors by price band to find weak spots: luxury homes (**13%** error vs. **6.5%** mid-priced) and low-sales towns.

Guang Labs | Engineering Intern | Remote **May 2026 - Aug 2026**

- Engineered a service for an AI video startup that classifies videos and images by type (sports, gaming, talking) using Gemini, retrying uncertain or malformed responses on different clips, up to **10** attempts.
- Stored each label, confidence score, and reasoning in a **PostgreSQL** table for auditing, backed by **86** tests.
- Developed an in-app notification bell that alerts users when their video finishes processing or fails.

Projects

Climate Change Hypothesis Testing | *Python, NumPy, Matplotlib* GitHub

- Ran an **A/B test** (permutation test, **5,000** simulations) to check whether drought years were truly drier than other years, rejecting chance at the 5% level ($p = \mathbf{0.026}$).
- Built **99%** bootstrap confidence intervals (**5,000** resamples) for monthly max temperatures, finding **9** of **12** months significantly hotter in 2019-2021 than in 1900-1960.
- Flagged the multiple-testing risk of running **12** tests at once at a **1%** cutoff.

LLM Arena Leaderboard | *Python, scikit-learn, pandas, Plotly*

- Ranked the top **20** LLMs from about **26K** human-preference battles by fitting a Bradley-Terry model (logistic regression) and comparing it to raw win rates on held-out battles.
- Bootstrapped **200** resamples for confidence intervals on each model's rank, and grouped prompts into topics with LSA and K-Means to compare leaderboards by topic.

Hindsight (Cal Hacks 13.0) | *Claude, FastAPI, React, Supabase* GitHub

- Co-designed a transfer planner: a **FastAPI** backend that matches courses to official ASSIST agreements with deterministic rules (minimum grades, agreement years), a **React** and **TypeScript** interface, and **Supabase**.
- Tested with **10** transfer students, who said it would have helped during their own transfer.

Leadership & Involvement

UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA **Aug 2026 - Present**

- Run exam reviews and write practice worksheets for CS and data science classes for transfer students, and give resume feedback.

UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.